

Multi-digit recognition, reading and writing

Unit one · more number of digits · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5, Volume A, Unit 1 + Modern Education, Volume 5, Unit 1

Core traps: □ T1 unit conversion trap - reading rules of 0 · Continuous carry domino effect · Omission of digit zero padding

SSPA association: ● High frequency Multi-digit reading and writing and approximate values must be tested every year

Prerequisite knowledge: Understand the reading rules of 0 ③ Be proficient in the "four digits and one stroke" sectioning method

Our goals: ① Master the reading and writing of eight/nine digits ② Understand the reading rules of 0 ③ Be proficient in the "four digits and one digit" segmentation method

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space
1	Write the number: Thirty-five thousand six hundred and seventy-two =?		
2	Read the number: 50,208 = ?		
3	In the number 8,406,125, where is "4"? How much does it mean?		
4	Write the number: seven million and five hundred =?		
5	Comparing sizes: Which is larger, 7,654,321 or 7,645,321?		

II.Core Knowledge + Worked Examples

Knowledge Point 1: Multi-digit numbers and reading and writing rules SSPA

- ① **Digit sequence (from right to left):** Units → Tens → Hundreds → Thousands → Tens of Thousands → Hundreds of Thousands → Millions → Tens of Tens → Hundreds of Hundreds
- ② **Hong Kong sectioning method (four digits per digit):** From right to left, draw one digit for every four digits: one level | ten thousand level | billion level
- ③ **Reading rules:** First read "Level 100" → then read "Level 10,000" → finally read "Level 1". Each level is internally read as thousands and hundreds of ten
- ④ **The reading method of 0 - the most important rule:** The 0 at the end of each level is not read; the consecutive 0s in the middle of each level only read one "zero"

Example of trap detonation (the most important demonstration in this class)

Read the number: 706,850,041

Common mistakes

706,850,041

Read the extra zeros!

Correct pronunciation

706,850,041

The 0 at the end of level 10,000 is not read, and the 0 in the middle of each level is only read as one.

 **Formula: "Four digits are divided into three levels, and billions of levels are read sequentially. There is no zero at the end of each level, and one in the middle is read continuously."**

 **The three most common reading errors:** ① The 0 at the end of each level is read out ② Multiple consecutive 0s are read multiple times in the middle ③ Forgetting to separate sections and reading directly

Knowledge Point - Worked Examples

#	Question	Difficulty	Working Space
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Example 1	Readout: 800,695,000		
Example 2	Write: five hundred and three thousand and sixty =?		

Knowledge Point 2: Digital Values and Number Composition ● SSPA

- ① Each number represents a different value at different **digits** ② For example, in 821,675,439: "8" in billion digits → **represents 800,000,000**; "2" in |||SEP||| → means 20,000,000 **Killer question type**: "Use numbers to form the largest/smallest number that meets the conditions" (the position rule of 0 must be considered)
- ③ **Killer question type**: "Use numbers to form the largest number/smallest number that meets the conditions" (the position rule of 0 must be considered)

example

Example 3: In 5,082,701, where is "8"? How much does it mean?

Knowledge point 3: Comparison and sorting of multi-digit numbers ● SSPA

- ① **Rules for comparing sizes**: Compare digits first (more digits → larger); same digits → compare digits one by one starting from the highest digit
- ② **Notes on sorting**: Make sure the question requires "from large to small" or "from small to large"
- ③ **Traps**: Numbers containing 0 are easily misjudged (such as 5,030,000 vs 5,003,000)

example

Example 4: Arrange 5,713,941, 57,168,931 and 5,719,871 from largest to smallest.

Knowledge Point 23 Synchronous practice

#	Question	Difficulty	Working Space
6	How much does "4" represent in 94,350,028? How much does "3" mean?		
7	Arrange the following numbers from largest to smallest: 5,300,000 / 5,030,000 / 5,003,000 / 5,000,300		
8	Readout: 500,055,005		
9	Write: 300,605,000 = ?		
10	Use the numbers 3, 0, 0, 5, 0, 5 to form a minimum six-digit number that "reads only one zero".		

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
11	Readout: 987,654,321		
12	Write: 654,321,000 =?		
13	Write: forty million and four hundred =?		
14	Which digits are 0 in 602,040,008?		
15	Write it in Arabic numerals: seven hundred and eight million and a half thousand =?		

Advanced layer (total 4 questions, choose do)

#	Question	Difficulty	Working Space
16	Write an eight-digit number without reading a single zero (each digit is different)		
17	Use 2, 0, 0, 7, 0, 0, 5, 1 to form the largest eight-digit number and the smallest eight-digit number		
18	An eight-digit number whose sum is 5, with each digit being different. What is the maximum?		
19	Use 3, 0, 0, 5, 0, 5, 0, 5 to form the maximum six-digit number that "reads only one zero".		
20	For an eight-digit number, its tens of millions and hundreds of thousands digits are both 0, and the other digits are different. If this number reads "not even a zero", what is the maximum number that can be read?		

applicationquestion (total 3 question, SSPAcommon situation)

#	Question	Difficulty	Working Space
21	Hong Kong has a population of approximately 7,413,000 people. Write this number in Chinese pronunciation.		
22	One YouTube channel has 104M views. What does M stand for? How much is it equal to?		
23	The government reported that "about 5,680,000 tourists visited Hong Kong". Is this number exact or approximate? Read it out.		
24	One housing estate in Hong Kong has 12,500 units and another housing estate has 8,750 units. Approximately how many units are there in the two housing estates? (First approximate the value to thousands and then calculate)		

IV. The ultimate challenge

#	Question	Difficulty	Working Space
 1	A nine-digit number, its tens of millions and millions digits are all 0, the numbers in other digits are different, and the sum of all numbers is 30. What is the largest possible number? What about the smallest one?		
 2	Use the numbers 0-9 one at a time to form a tens digit (0 cannot be in the highest position). Requirement: When reading out, the number of "zeros" should be the minimum. How can this number be written? How many times do you read zero at least?		
 3	There is a nine-digit number, its billion digit is 2, the tens of millions and millions digits are both 0, and the hundred thousand digit is 5. If the number reads only two "zeros", the remaining numbers are different and as large as possible. What is this number? Read it out.		

V. Class afterhomework

Basic must-do (total 6 questions)

#	Question	Difficulty	Working Space
H1	Readout: 350,000,000		
H2	Write: 987,654,321 =?		
H3	Write: fifty million and five		
H4	Which digits are 0 in 405,060,700?		
H5	Arrange 2,345,678 and 2,346,578 and 2,354,678 from largest to smallest		
H6	Write out: one hundred and twenty-three million four hundred and fifty-six thousand seven hundred and eighty-nine =?		

For advanced, choose do (total 2 questions, choose do)

#	Question	Difficulty	Working Space
H7	Use 1, 0, 0, 0, 2, 0, 0, 3 to form the smallest eight-digit number that "reads two zeros"		
H8	In an eight-digit number, the sum of each adjacent three-digit number is 10. What might this number be? (Give two examples)		
H9	Use the numbers 1-9 one at a time to form a nine-digit number. Requirement: When reading out, the number of "zero" is the most. How can this number be written? How many times can you read zero at most?		

VI. The Lessoncorecommon errorssummary

 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve 🌱 basic questions independently (100% correct)

☐ Challenge 🌿 Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Sectioning error SEP : three characters and one word (mainland custom): Three digits and one digit (mainland custom)	Hong Kong uses four digits: one, ten thousand, and billions
2	The 0 at the end of each level reads out SEP : 806.95 million: 806.95 million	The 0 at the end of each level does not read: 806.95 million
3	Read multiple times with 0 in the middle	Continuous 0 reads only one "zero"
4	Fill in missing zeros when writing numbers SEP : fifty million and five → 50000005: 50 million and five→50000005	Correct: 50,000,005 (0 should be added to the middle seven digits)
5	Ratio size ignores the effect of 0	Bit by bit comparison, 0 is also a number